

CLOUDWISE AI: AI-POWERED FINOPS PREDICTIVE CLOUD COST OPTIMIZER

ALVIN E Y
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Mini Project

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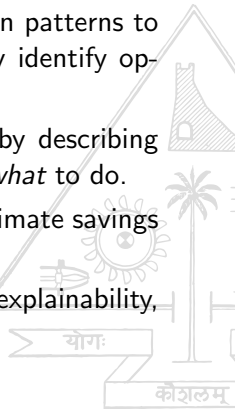
Introduction

- Cloud computing is the preferred platform for modern applications due to scalability, flexibility, and pay-as-you-go pricing.
- Organizations often incur unnecessary cloud expenditure from idle virtual machines, oversized instances, unused storage volumes, and always-on development environments.
- As cloud infrastructures grow more complex, manually identifying such inefficiencies becomes increasingly difficult for administrators.
- Existing cloud cost management tools mainly monitor usage and give basic, rule-based recommendations – without prediction, explanation, or risk assessment.



Introduction

- Machine Learning can analyze historical utilization patterns to forecast future cloud expenditure and proactively identify optimization opportunities.
- Explainable AI (XAI) builds administrator trust by describing *why* a recommendation was generated, not just *what* to do.
- A What-If Cost Simulation lets administrators estimate savings and risk before touching live infrastructure.
- CloudWise AI integrates monitoring, prediction, explainability, and simulation into one FinOps assistant.





Introduction

Objective:

- Develop **CloudWise AI**, an Explainable AI-based FinOps Assistant, to help organizations monitor, manage, and reduce cloud expenditure.
- Predict future cloud expenditure using accurate and explainable forecasting models.
- Generate risk-aware and explainable cloud cost-optimization recommendations.
- Provide interactive *What-If* cost simulations to support scenario planning and informed decision-making.
- Reduce unnecessary cloud costs through transparent, data-driven decision support.



Literature Review

Sutar, P. and Gunjan, V., "AI-Driven Multi-Objective Resource Scheduling for Cost and Performance Optimization in Dynamic Cloud Computing Environments," *Int. J. Advanced Computer Science and Applications*, vol. 17, no. 2, 2026.

- Combines LSTM-based workload forecasting, Isolation Forest anomaly detection, and multi-objective scheduling to optimize cost while maintaining SLA compliance.
- Demonstrates significant cost reduction and improved resource utilization.
- **Observation:** Focuses on scheduling/allocation; lacks explainable recommendations, risk assessment, and pre-deployment simulation.



Literature Review

Bhaskaran, S. et al., "AI-Driven Cloud Optimization: Enhancing Cost Prediction, Resource Scheduling and Fault Resilience in Cloud Environments," *IEEE CloudCom*, 2025.

- Integrates ML-based cost prediction, Deep Reinforcement Learning scheduling, and Isolation Forest anomaly detection in a unified dashboard.
- Improves cost estimation, scheduling, and fault resilience.
- **Observation:** No demand forecasting for rightsizing, and lacks risk evaluation, explainability, or simulation capabilities.



Literature Review

Zambianco, M. et al., “Cost Minimization in Multi-cloud Systems with Runtime Microservice Re-orchestration,” *IEEE ICIN*, 2024.

- Uses Integer Linear Programming and heuristics to migrate microservices across cloud providers, minimizing cost while maintaining QoS.
- Rolling-update mechanism reduces service disruption during migration.
- **Observation:** Focuses on runtime orchestration/migration – no forecasting, recommendation, explainability, or risk-aware support.



Literature Review

Basu, D. et al., “DRIVE: Dynamic Resource Introspection and VNF Embedding for 5G Using Machine Learning,” *IEEE Internet of Things Journal*, vol. 10, no. 21, 2023.

- Uses ML and optimization for intelligent Virtual Network Function allocation in software-defined 5G networks.
- Improves resource utilization and reduces infrastructure cost in the NFV domain.
- **Observation:** Targets network function virtualization, not cloud cost optimization; no forecasting, rightsizing, XAI, or risk analysis.



Literature Review

Zhang, Q. et al., “A Cost-Effective Hybrid Cloud Resource Scaling Framework for Batch Processing Services,” *IEEE Trans. Network Science and Engineering*, vol. 12, no. 1, 2025.

- Reinforcement-learning-based hybrid VM/FaaS scaling that adjusts resources to workload while minimizing cost.
- Improves cost efficiency over conventional scaling techniques for batch workloads.
- **Observation:** Reactive automatic scaling only – no predictive rightsizing, explainability, or pre-modification risk analysis.



Literature Review

Author	Approach	Brief Observation
Sutar & Gunjan (2026)	LSTM+scheduling	Forecasting-driven scheduling; no explainability or risk.
Bhaskaran et al. (2025)	ML cost prediction+DRL	Accurate prediction; lacks right-sizing forecast & simulation.
Zambianco et al. (2024)	ILP microservice re-orchestration	Minimizes cost via migration; no recommendation support.
Basu et al. (2023)	ML-based VNF allocation	NFV-focused; not cloud cost optimization.
Zhang et al. (2025)	RL hybrid VM/FaaS scaling	Reactive scaling; no predictive rightsizing or risk analysis.



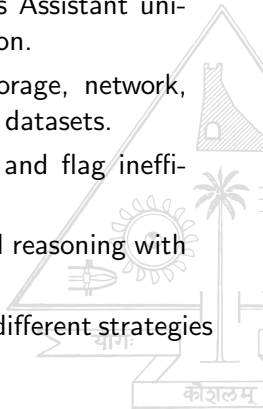
Existing System

- Cloud providers and third-party FinOps platforms offer dashboards showing utilization metrics, billing reports, and expenditure summaries.
- Common features: idle VM detection, VM rightsizing recommendations, budget alerts, cost allocation reports, and basic anomaly detection/forecasting.
- These systems mainly rely on predefined rules and static threshold-based analysis.
- Administrators must still manually evaluate recommendations before implementation.
- **Key limitations:** rule-based (not AI-driven) decisions; limited future-cost prediction; recommendations without explanation; no operational risk estimation; no What-If simulation; no Cloud Efficiency Score or FinOps maturity assessment.



Proposed System

- CloudWise AI: an Explainable AI-based FinOps Assistant unifying monitoring, prediction, XAI, and simulation.
- **Data Collection:** Gathers CPU, memory, storage, network, and billing metrics via cloud APIs or simulated datasets.
- **Prediction:** ML models forecast expenditure and flag inefficient allocation.
- **Explainable Recommendations:** SHAP-based reasoning with savings and risk score.
- **What-If Simulation:** Estimates savings under different strategies without touching live infrastructure.

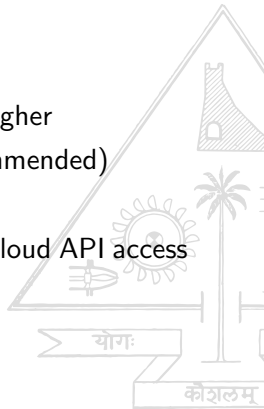




Requirements Specification

Hardware Requirements

- **Processor:** Intel Core i5 / AMD Ryzen 5 or higher
- **Memory:** Minimum 8 GB RAM (16 GB recommended)
- **Storage:** Minimum 100 GB free disk space
- **Internet Connectivity:** Stable broadband for cloud API access





Requirements Specification

Software Requirements

- **Operating System:** Windows 10/11 / Ubuntu 22.04 LTS / macOS
- **Programming Language:** Python 3.10 or later
- **Frontend:** React.js with HTML5, CSS3, JavaScript
- **Backend:** FastAPI (REST APIs)
- **Database:** PostgreSQL / MongoDB
- **Cloud SDK:** AWS Boto3 SDK (or simulated cloud datasets)
- **ML Libraries:** Scikit-learn; Prophet / XGBoost for forecasting; SHAP for explainable AI





Requirements Specification

- **IDE Used:** Visual Studio Code
- **Version Control:** Git and GitHub
- **Containerization:** Docker (optional)
- **Visualization:** Chart.js / Plotly





Requirements Specification

Functional Requirements

- **User Interface:** Interactive dashboard for connecting cloud accounts or uploading simulated datasets; visualizes metrics, expenditure, recommendations, and predictions.
- **Cloud Data Collection:** Gathers CPU, memory, storage, network, and billing data from provider APIs or simulated datasets, with validation for completeness and consistency.
- **Cloud Resource Monitoring:** Continuously observes utilization across VMs, storage, databases, and networking to identify inefficient allocation.
- **Predictive Cost Analysis:** ML models forecast future cloud expenditure from historical utilization trends.



Requirements Specification

- **Explainable Recommendation Engine:** Produces resource name, current utilization, expected monthly savings, XAI-driven explanation, confidence score, and risk level for each recommendation.
- **Risk Assessment:** Classifies optimization actions into Low, Medium, or High risk using utilization patterns and resource dependencies.
- **What-If Cost Simulation:** Estimates expenditure after applying different strategies without modifying the live environment.
- **Dashboard & Report Generation:** Presents utilization, predicted expenditure, optimization opportunities, and Cloud Efficiency Scores; generates comprehensive reports.



Requirements Specification

Non-Functional Requirements

- **Scalability:** Handles small to large cloud infrastructures; extensible to AWS, Azure, GCP.
- **Usability:** Intuitive dashboard requiring no advanced technical knowledge.
- **Reliability:** Consistent, accurate reports with proper exception handling.
- **Performance:** Efficient processing with real-time dashboard visualization.
- **Security:** Encrypted communication and secure authentication for cloud credentials and billing data.
- **Maintainability:** Modular architecture simplifying integration of new ML models and cloud providers.



Input and Output

- **Input:** Cloud resource utilization and billing data, obtained either by connecting a live cloud account (e.g. via AWS Boto3) or by uploading a simulated/historical usage dataset.
- **Input Format:** CPU, memory, storage, network, and billing metrics retrieved as structured API responses (JSON), or historical usage and cost records supplied as CSV/dataset files.
- **Output:** Predicted future cloud expenditure, explainable optimization recommendations (with estimated savings, confidence score, and risk level), What-If simulation results, and a Cloud Efficiency Score.
- **Output Format:** Interactive dashboard visualizations, SHAP-based explanation summaries, and downloadable reports presenting predictions, recommendations, and simulation outcomes.



Conclusion

- CloudWise AI integrates cloud monitoring, ML-based cost prediction, Explainable AI, and What-If simulation into a unified FinOps assistant.
- Unlike existing rule-based platforms, it forecasts future expenditure, explains every recommendation, and quantifies operational risk before action is taken.
- Aims to reduce unnecessary cloud expenditure while improving administrator confidence and transparency in AI-assisted decisions.
- Designed using open-source technologies and simulated cloud datasets, making it a feasible, extensible FinOps decision-support platform.



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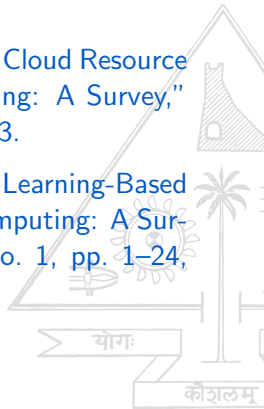
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Thank You

